

## LOW TGP RTX 4050 GAMING LAPTOPS

**MSI Thin GF63 12VE**

Two of our contenders in this category come from MSI, with the first being the MSI Thin GF63. The Thin lineup is MSI's budget gaming lineup and as such comes with weaker TGP GPUs. However, costing a little over a lac, this MSI Thin GF63 still comes with a pretty decent processor in the 12th gen Intel Core i7-12650H, albeit with only 8GB of DDR4 RAM.

**Specs:**

**GPU:** NVIDIA GeForce RTX 4050 Laptop GPU | **TGP:** 45W | **CPU:** Intel Core i7-12650H | **RAM:** 8GB DDR4 | **Price:** ₹1,09,862

**MSI Cyborg 15 A12VE**

The MSI Cyborg lineup is new from MSI having started only this year, and sits somewhere in between the MSI Thin and Katana lineup. You have variants with both high and low TGP and more powerful GPU offerings as well. However, the variant we've tested features a TGP of 45W, like the Thin GF63. You have the same processor as well, in the 12th gen Intel Core i7-12650H, however, unlike the Thin you have 16GB of DDR5 RAM in the Cyborg. Considering the price it's pretty clear on paper which is the better choice.

**Specs:**

**GPU:** NVIDIA GeForce RTX 4050 Laptop GPU | **TGP:** 45W | **CPU:** Intel Core i7-12650H | **RAM:** 16GB DDR5 | **Price:** ₹99,990

**Dell G15 5520 (RTX 3050)**

Finally, we're going to be comparing the two low TGP RTX 4050 gaming laptops with an RTX 3050 Laptop, the Dell G15 5520. Unlike the other two laptops we've got a 12th gen Intel Core i5-12500H here, which in terms of performance isn't really that far behind the 12650H. Other than that, you also have 16GB of DDR5 RAM and a much higher TGP of 95W on the RTX 3050 on-board.

**Specs:**

**GPU:** NVIDIA GeForce RTX 3050 Laptop GPU | **TGP:** 95W | **CPU:** Intel Core i5-12500H | **RAM:** 16GB DDR5 | **Price:** ₹72,990

stays within a generation; a generational leap can potentially result in significant performance improvements, though it may not always be true. Enough speculation, we've got the benchmark results right here so we can see for ourselves.

**Synthetic GPU – Low TGP RTX 4050 vs RTX 3050**

We use the 3DMark benchmark to measure synthetic performance; Time Spy for DirectX12 performance and Fire Strike for DirectX11 performance tests.

As you can see in the graph, both the RTX 4050 laptops perform better than the RTX 3050 in every benchmark. Unsurprisingly, the MSI Cyborg manages to outperform the MSI Thin GF63 thanks to its RAM having both higher capacity and much faster speeds.

16GB of RAM can really make a difference when compared to 8GB of RAM, especially when it comes to gaming. The fact that it's DDR5 over DDR4 also helps.

**Real-world Gaming – Low TGP RTX 4050 vs RTX 3050**

However, synthetic GPU performance may not necessarily translate to real-world performance. This is where our real-world gaming benchmarks come in. You'll notice that both the RTX 4050 GPU laptops are ahead in most games, but the RTX 3050 really isn't all that far behind. The RTX 3050 even outperforms the 45W RTX 4050s in some titles.

You'll notice that the games where the RTX 3050 outperforms the RTX 4050 are older, which aren't as optimised for newer GPUs. DirectX11

tends to give better performance on RTX 30 series cards, while the same is true for DirectX12 games and RTX 40 series cards. However, newer and more graphic intensive titles will run better on the RTX 4050 in this case, even though they have a lower TGP. While this low TGP comparison might still leave some doubts about whether or not you should go for an RTX 4050 laptop, our next batch will definitely quash all of them.

**High TGP RTX 4050 Gaming Laptops**

We've got a total of 5 laptops for this comparison, with 4 of them featuring an RTX 4050 with the lowest TGP being 120 W. The 5th laptop features an RTX 3060, to show you how much of a performance difference there is just within RTX 4050s how much difference TGP actually makes. With that, let's get to our next contenders (*table on the next page*).

With all 5 candidates on the table, let's immediately get to the benchmarks.

